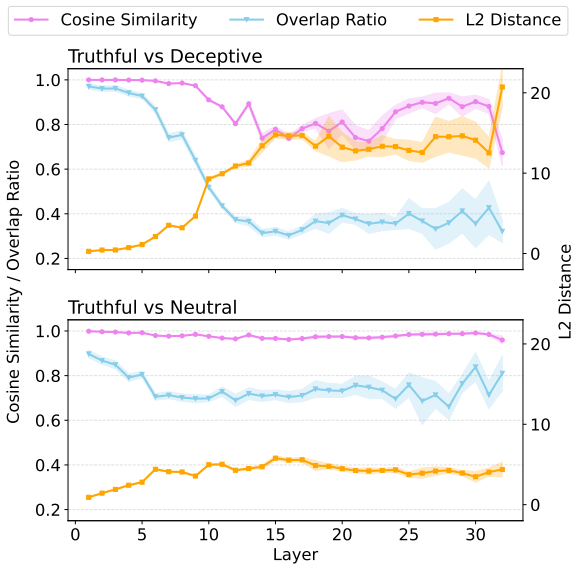


Figure 6: SAE-based analysis: Layer-wise feature shift analysis for LLaMA-3.1-8B-Instruct on `cities`. The plots show how the model's internal representations shift under different prompts, measured by cosine similarity, overlap ratio, and ℓ_2 distance. The top panel (Truthful vs. Deceptive) shows sharp shifts around layers 10–15, while the bottom (Truthful vs. Neutral) shows smaller but consistent changes. Shaded regions show $\pm 1\sigma$ across samples.

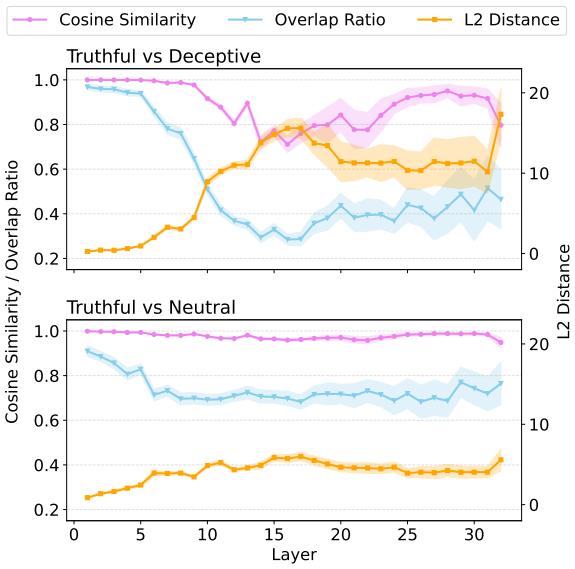


Figure 7: SAE-based analysis: Layer-wise feature shift analysis for LLaMA-3.1-8B-Instruct on `common_claim_true_false`.

most responsive sparse features.

3.2 Representational Geometry

We now test whether the peak layers identified by probing also expose the clearest geometric split under different instructions.

3.2.1 PCA Separation on Curated vs. Complex Data

For the curated `cities` set, a 2-D PCA of LLaMA-3.1-8B activations cleanly pulls apart TRUE and FALSE samples under all three prompts: the clusters begin to split layers 8, are almost lienarly separable by layer 14, and remain distinct through layer 32 (Figure 4). These are exactly the depths where linear-probe accuracy peaks. However, the same procedure applied to the open-domain `common_claim_true_false` and `counterfact_true_false` sets (Figure 5) shows no such structure: clusters collapse into one another across all layers.

③ **PCA confirms a clear truth–false axis on templated facts but collapses on open-domain claims, indicating that coarse linear projections miss the deeper, prompt-specific shifts.** Projecting a 4 k–5 k dimensional residual vector onto two principal components preserves only the directions of greatest *global* variance; in longer sentences those directions are dominated by lexical and syntactic variation. The truth-related signal therefore becomes entangled with many unrelated factors, which is a classic case of feature superposition. Thus, the clusters flatten into an indistinct cloud.

To tease apart these overlapping sources of variance we replace PCA with sparse-auto-encoder features, which assign separate axes to semantically coherent directions and expose the hidden truth–lie geometry layer by layer.

3.2.2 SAE Feature Shifts Quantify Geometry

Figures 6 and 7 track three layer-wise distances between the **truthful** centroid and its **deceptive** or **neutral** counterpart on LLaMA-3.1-8B-instruct. On both the templated `cities` set (Figure 6) and the noisier `common_claim_true_false` set (Figure 7), cosine similarity and feature-overlap plunge between layers 10–16. Meanwhile, the ℓ_2 distance climbs to a clear peak. Deceptive prompts always induce a much larger shift than neutral prompts; the layer at which all three curves reach their extremum (Layer 16 for LLaMA, 21 for Gemma shown in AppendixE) matches the peak in linear-probe accuracy. In contrast, the curves for *truthful vs. neutral* stay almost flat, with cosine > 0.95 and overlap > 0.80 throughout.

④ **The geometric pattern of feature shifts is consistent regardless of dataset complexity, con-**

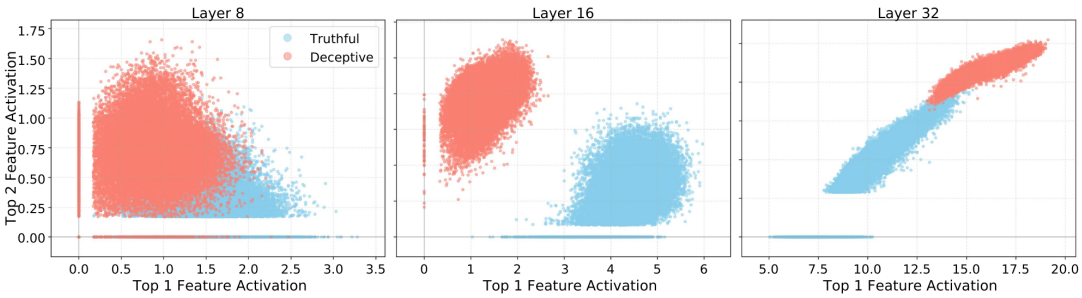
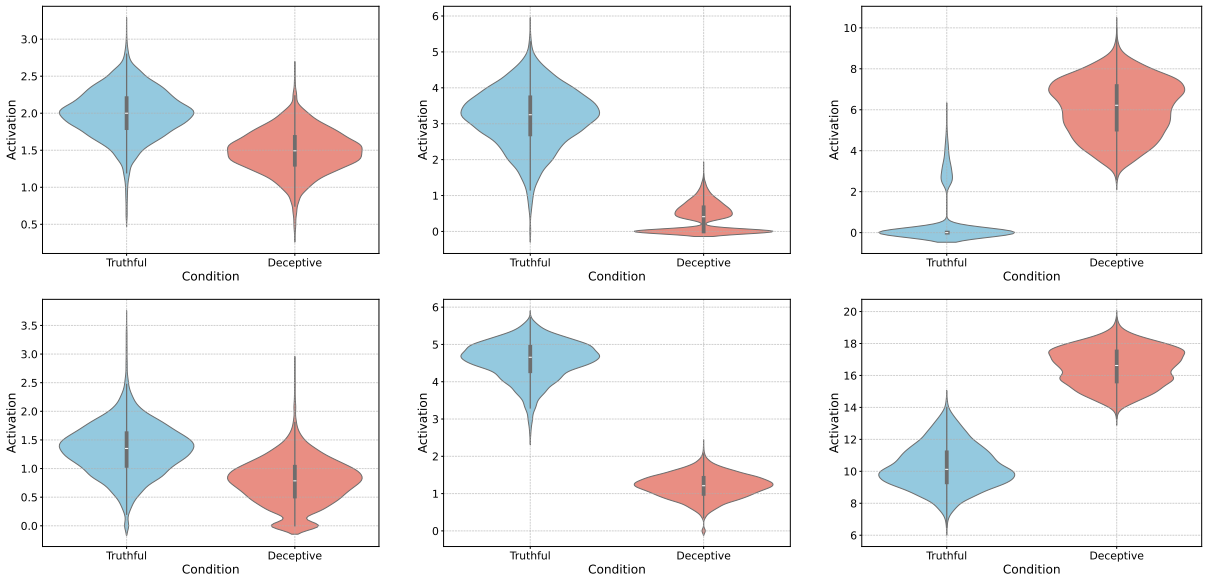


Figure 8: Neuron-level SAE feature shifts on `CommonClaim` (LLaMA-3.1-8B-Instruct). At layers 8, 16, and 32, we show mean SAE activations under *Truthful* vs. *Deceptive* for the two most deception-sensitive features in that layer (left/right; ranked by $\Delta_i = |\bar{f}_{\text{decep},i} - \bar{f}_{\text{truth},i}|$).



(a) Layer 8 (b) Layer 16 (c) Layer 32

Figure 9: Violin graph of LLaMA3.1-8B-Instruct activations on the `common_claim_true_false` dataset. The top row displays the activation distributions for the SAE feature most responsive to deceptive instructions (Top 1 feature), while the bottom row shows the distributions for the second most responsive feature (Top 2 feature), across layers 8 (a), 16 (b), and 32 (c).

firming a stereotyped truth–lie reorientation rather than dataset-specific noise. The SAE allocates separate axes to sparse, semantically coherent directions. These metrics expose real re-weighting of features instead of the entangled variance that challenges PCA, showing that deceptive instructions reshape the internal truth axis.

3.2.3 Neuron-by-Neuron Analysis: Key Sparse Features Flip Sign

While SAE feature shifts reveal robust geometric differences under different prompt types, they do not explain which specific neurons are responsible for these shifts. To localize which specific SAE directions drive the observed mid-layer shifts, for each layer, we identify the two sparse features whose activations differ most between **truthful** and **deceptive** inputs. Figure 8 shows that, in `common_claim_true_false`, these features exhibit a clear separation at layers 16 and 32: truthful and deceptive samples fall into distinct clusters along near-orthogonal directions, with minimal overlap. Similar trends are observed for another uncurated dataset `counterfact_true_false` (Appendix E). Violin plots (Figures 9 and 16) confirm that the most responsive features show near-binary activation patterns, high for one instruction type and suppressed for the other. For example, top 2 features in Layer 16 is active almost exclusively under truthful prompts, while in Layer 32 this flips, activating strongly for deceptive inputs but not truthful ones.

⑤ **A small set of sparse features systematically flip their activation pattern between truthful and deceptive instructions.** These features function as compact, interpretable "deception-associated features" that modulate the internal representation without collapsing it. Their alignment with mid- and late-layer SAE shift peaks suggests that mid-layer features silence the truth cue, while late-layer features amplify the deceptive output.

4 Related Work

Early studies showed that truth-related signals are encoded in activations and can be decoded via probes (Azaria and Mitchell, 2023a; Liu et al., 2024; Jin et al., 2025). Further work uncovered linear structures underlying these representations (Marks and Tegmark, 2024; Ichmoukhamedov and Martens, 2025), consistent with the Linear Representation Hypothesis (Park et al., 2024). Various probing techniques, from Logistic Regression (LR) (Li et al., 2024; Marks and Tegmark, 2024) to polarity-aware approaches like TTPD (Bürger et al., 2024), have been used to find these "truth directions", although generalization remains a challenge (Marks and Tegmark, 2024; Bürger et al., 2024). Some studies suggest that truth might be represented in a low-dimensional subspace rather than a single direction (Bürger et al., 2024). Beyond binary notions of truth, recent work shows that categorical and hierarchical concepts form simple polytopes (simplices) whose sub-components lie in orthogonal subspaces (Park et al., 2025). Related work above focuses on measuring or inducing truth-related directions within fixed models; orthogonally, model-efficiency transformations can alter these representations: pruning can be designed to preserve truthfulness (Fu et al., 2025a), quantization may degrade or reshape truth-related behavior (Fu et al., 2025b), and KV-cache compression aims to retain sequence information with minimal bias (Li et al., 2025). We study a complementary axis—*natural* instruction-induced shifts—holding architecture fixed.

Instruction-following behavior has also been linked to internal states (Heo et al., 2025), with Representation Engineering (Zou et al., 2025) and related methods demonstrate showing causal control over outputs (Li et al., 2024; Marks and Tegmark, 2024), including knowledge-based refusals (Ferrando et al., 2025). Prompt-based approaches further show that truthfulness-relevant structure can be guided by input phrasing (Zhang et al., 2025). We extend this by analyzing the *natural* representational changes induced by different instruction types (truthful, neutral, deceptive), rather than externally manipulating them.

Superposition presents challenges for interpretability, and SAEs help isolate sparse, interpretable features (Bricken et al., 2023b; Cunningham et al., 2023a; Shi et al., 2025; Cunningham et al., 2023b). Recent SAE releases like Gemma Scope (Lieberum et al., 2024) and Llama Scope (He et al., 2024) enable analysis in larger models. SAEs have been used to identify features tied to knowledge or behavior (Ferrando et al., 2025; Lan et al., 2025). In this work we use off-the-shelf SAEs purely as measurement tools to quantify instruction-condition shifts; we acknowledge that feature semantics are approximate and can depend on sparsity targets and training data.

5 Conclusion

This paper explored how deceptive instructions alter the internal representational geometry of LLMs compared to truthful or neutral ones. We found that the model's instructed "True" or "Flase" output is linearly decodable from intermediate activations across instruction types and datasets. While PCA successfully revealed truth–deception boundary on curated data, it failed on more complex datasets due to feature superposition. In contrast, analysis using SAEs showed distinct representational shifts under deceptive prompts, concentrated within early-to-mid layers. A neuron-level analysis further identified a few sparse features with polarity flips, serving as interpretable "deception-associated features". These insights clarify the internal geometry of instructed dishonesty in LLMs and offer a solid basis for future deception detection and model editing methods.

Limitations

Our study is confined to English declaratives, frozen model weights, and linear probes. It neither tests causal interventions (e.g. activation patching) nor adversarial prompt recombinations. Furthermore, the evaluation data are labelled for binary factuality only; future work should extend to graded truth scales and multilingual settings.